

# DATA309 Housing Affordability — Statistical Analysis

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2026-10-07

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## 1. Analysis Overview

This analysis investigates housing affordability and housing-market conditions in Canterbury.

The analysis focuses on three questions:

1. How does housing affordability in Canterbury compare with New Zealand?
2. What economic and housing-market factors are associated with Canterbury house-price growth?
3. Have household incomes and rents kept pace with Canterbury house prices?

The analysis uses descriptive statistics, graphical exploratory analysis, paired tests, regression modelling, lagged explanatory variables and time-series diagnostics.

Because the data are observational, regression coefficients are interpreted as **associations rather than causal effects**.

---

## 2. Data Import

All datasets should be stored in the same project folder as this R Markdown file.

```
affordability_file <- first_existing(
  c(
    "house-value-to-income-multiple-and-mortgage-payment-proportion-of-income (1).csv",
    "house-value-to-income-multiple-and-mortgage-payment-proportion-of-income.csv"
  )
)

time_series_file <- first_existing(
  c(
    "Time_series_data.xlsx",
    "Time_series_data(1).xlsx",
    "Time_series_data(2).xlsx",
    "Time_series_data (1).xlsx",
    "Time_series_data (2).xlsx"
  )
)
```

```

)

affordability <- read.csv(
  affordability_file,
  stringsAsFactors = FALSE
)

# Standardise CSV column names
names(affordability) <- make.names(
  names(affordability)
)

ts_raw <- read_excel(
  time_series_file
)

cat(
  "Affordability file:",
  affordability_file,
  "\n"
)

```

## Affordability file: house-value-to-income-multiple-and-mortgage-payment-proportion-of-income (1).csv

```

cat(
  "Time-series file:",
  time_series_file,
  "\n"
)

```

## Time-series file: Time\_series\_data.xlsx

### 3. Analysis 1 — Canterbury vs New Zealand Affordability

#### 3.1 Prepare affordability data

```

affordability_compare <- affordability %>%
  transmute(
    Year =
      Year,

    Canterbury_ValueIncome =
      Canterbury.Region.House.value.to.income,

    NZ_ValueIncome =
      New.Zealand.House.value.to.income,

    Canterbury_Mortgage =
      Canterbury.Region.Mortgage.payment.proportion.of.income.,
  )

```

```

    NZ_Mortgage =
      New.Zealand.Mortgage.payment.proportion.of.income.
  ) %>%
  mutate(
    ValueIncome_Gap =
      Canterbury_ValueIncome -
      NZ_ValueIncome,

    Mortgage_Gap =
      Canterbury_Mortgage -
      NZ_Mortgage
  ) %>%
  arrange(Year)

kable(
  affordability_compare,
  digits = 2,
  col.names = c(
    "Year",
    "Canterbury Value-to-Income",
    "NZ Value-to-Income",
    "Canterbury Mortgage Burden (%)",
    "NZ Mortgage Burden (%)",
    "Value-to-Income Gap",
    "Mortgage Gap (pp)"
  ),
  caption =
    "Canterbury and New Zealand Affordability Measures"
)

```

Table 1: Canterbury and New Zealand Affordability Measures

| Year | Canterbury<br>Value-to-Income | NZ Value-<br>to-Income | Canterbury<br>Mortgage Burden<br>(%) | NZ Mortgage<br>Burden (%) | Value-to-<br>Income Gap | Mortgage<br>Gap (pp) |
|------|-------------------------------|------------------------|--------------------------------------|---------------------------|-------------------------|----------------------|
| 2005 | 4.4                           | 4.8                    | 31.8                                 | 34.8                      | -0.4                    | -3.0                 |
| 2006 | 4.8                           | 5.2                    | 35.0                                 | 37.7                      | -0.4                    | -2.7                 |
| 2007 | 5.1                           | 5.4                    | 40.0                                 | 42.6                      | -0.3                    | -2.6                 |
| 2008 | 4.8                           | 5.1                    | 40.5                                 | 43.6                      | -0.3                    | -3.1                 |
| 2009 | 4.3                           | 4.7                    | 27.2                                 | 29.5                      | -0.4                    | -2.3                 |
| 2010 | 4.5                           | 4.9                    | 30.9                                 | 33.8                      | -0.4                    | -2.9                 |
| 2011 | 4.2                           | 4.6                    | 27.4                                 | 29.8                      | -0.4                    | -2.4                 |
| 2012 | 4.3                           | 4.6                    | 26.2                                 | 28.0                      | -0.3                    | -1.8                 |
| 2013 | 4.7                           | 5.0                    | 27.6                                 | 29.3                      | -0.3                    | -1.7                 |
| 2014 | 4.7                           | 5.1                    | 29.8                                 | 32.2                      | -0.4                    | -2.4                 |
| 2015 | 4.8                           | 5.4                    | 29.5                                 | 33.3                      | -0.6                    | -3.8                 |
| 2016 | 4.9                           | 5.8                    | 27.7                                 | 33.0                      | -0.9                    | -5.3                 |
| 2017 | 5.0                           | 6.4                    | 28.2                                 | 36.2                      | -1.4                    | -8.0                 |
| 2018 | 4.9                           | 6.4                    | 27.4                                 | 36.2                      | -1.5                    | -8.8                 |
| 2019 | 4.8                           | 6.2                    | 26.3                                 | 34.3                      | -1.4                    | -8.0                 |
| 2020 | 4.8                           | 6.3                    | 24.7                                 | 32.9                      | -1.5                    | -8.2                 |
| 2021 | 5.5                           | 7.5                    | 26.2                                 | 36.1                      | -2.0                    | -9.9                 |

| Year | Canterbury<br>Value-to-Income | NZ Value-<br>to-Income | Canterbury<br>Mortgage Burden<br>(%) | NZ Mortgage<br>Burden (%) | Value-to-<br>Income Gap | Mortgage<br>Gap (pp) |
|------|-------------------------------|------------------------|--------------------------------------|---------------------------|-------------------------|----------------------|
| 2022 | 6.5                           | 8.0                    | 36.8                                 | 45.0                      | -1.5                    | -8.2                 |
| 2023 | 5.8                           | 6.6                    | 39.6                                 | 44.8                      | -0.8                    | -5.2                 |
| 2024 | 5.7                           | 6.4                    | 40.2                                 | 44.7                      | -0.7                    | -4.5                 |
| 2025 | 5.6                           | 6.1                    | 34.0                                 | 36.8                      | -0.5                    | -2.8                 |
| 2026 | 5.6                           | 5.9                    | 33.1                                 | 35.0                      | -0.3                    | -1.9                 |

### 3.2 Average affordability

```
comparison_summary <- affordability_compare %>%
  summarise(
    Canterbury_Mean_ValueIncome =
      mean(
        Canterbury_ValueIncome,
        na.rm = TRUE
      ),

    NZ_Mean_ValueIncome =
      mean(
        NZ_ValueIncome,
        na.rm = TRUE
      ),

    Mean_ValueIncome_Gap =
      mean(
        ValueIncome_Gap,
        na.rm = TRUE
      ),

    Canterbury_Mean_Mortgage =
      mean(
        Canterbury_Mortgage,
        na.rm = TRUE
      ),

    NZ_Mean_Mortgage =
      mean(
        NZ_Mortgage,
        na.rm = TRUE
      ),

    Mean_Mortgage_Gap =
      mean(
        Mortgage_Gap,
        na.rm = TRUE
      )
  )

kable(
```

```
comparison_summary,
digits = 2,
col.names = c(
  "Cant. HVI",
  "NZ HVI",
  "HVI Gap",
  "Cant. Mortgage (%)",
  "NZ Mortgage (%)",
  "Mortgage Gap (pp)"
),
caption =
  "Average Affordability: Canterbury vs New Zealand"
)
```

Table 2: Average Affordability: Canterbury vs New Zealand

| Cant. HVI | NZ HVI | HVI Gap | Cant. Mortgage (%) | NZ Mortgage (%) | Mortgage Gap (pp) |
|-----------|--------|---------|--------------------|-----------------|-------------------|
| 4.99      | 5.75   | -0.76   | 31.37              | 35.89           | -4.52             |

### Interpretation

Across the available period, Canterbury's average house value-to-income multiple was approximately **4.99**, compared with approximately **5.75** nationally.

Canterbury's average mortgage-payment burden was approximately **31.37%**, compared with approximately **35.89%** nationally.

This suggests that Canterbury has generally been more affordable than New Zealand overall.

## 3.3 House value-to-income comparison

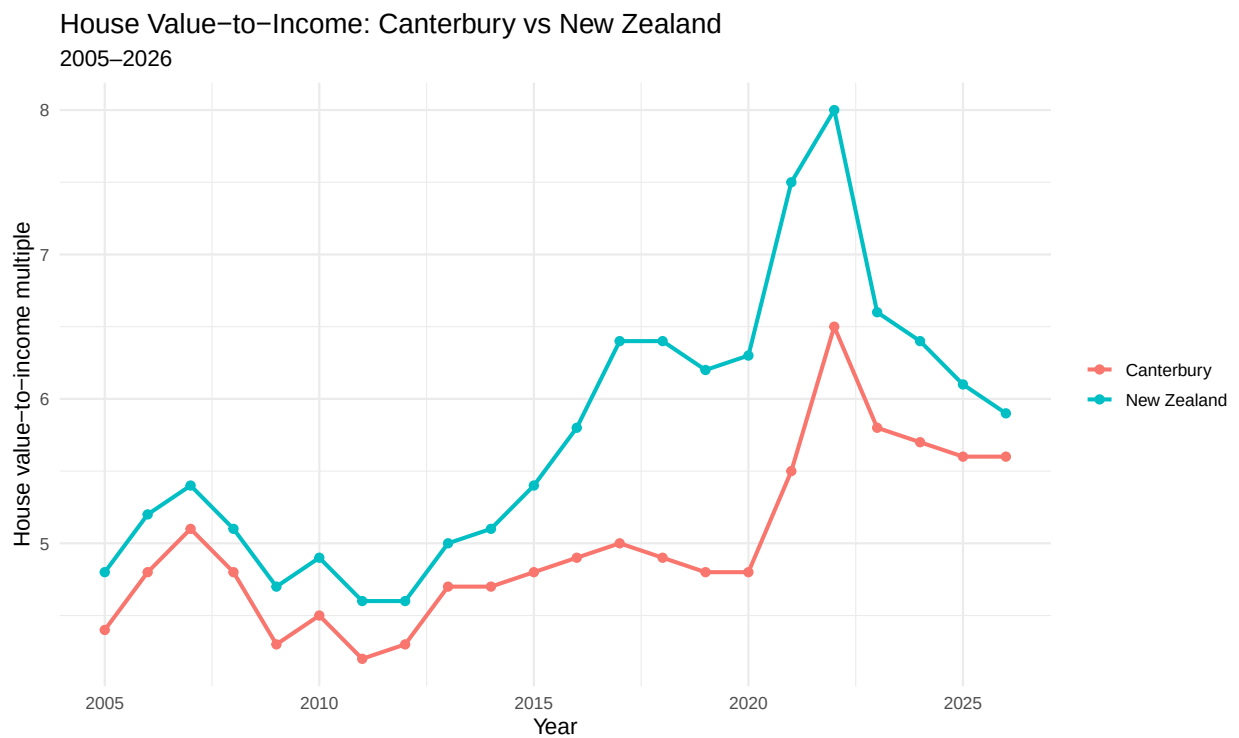
```
value_income_long <- affordability_compare %>%
  select(
    Year,
    Canterbury =
      Canterbury_ValueIncome,
    `New Zealand` =
      NZ_ValueIncome
  ) %>%
  pivot_longer(
    -Year,
    names_to = "Region",
    values_to = "ValueIncome"
  )

ggplot(
  value_income_long,
  aes(
```

```

    x = Year,
    y = ValueIncome,
    colour = Region
  )
) +
  geom_line(
    linewidth = 1.1
  ) +
  geom_point(
    size = 2
  ) +
  labs(
    title =
      "House Value-to-Income: Canterbury vs New Zealand",
    subtitle =
      "2005-2026",
    x = "Year",
    y =
      "House value-to-income multiple",
    colour = NULL
  ) +
  theme_minimal(
    base_size = 13
  )
)

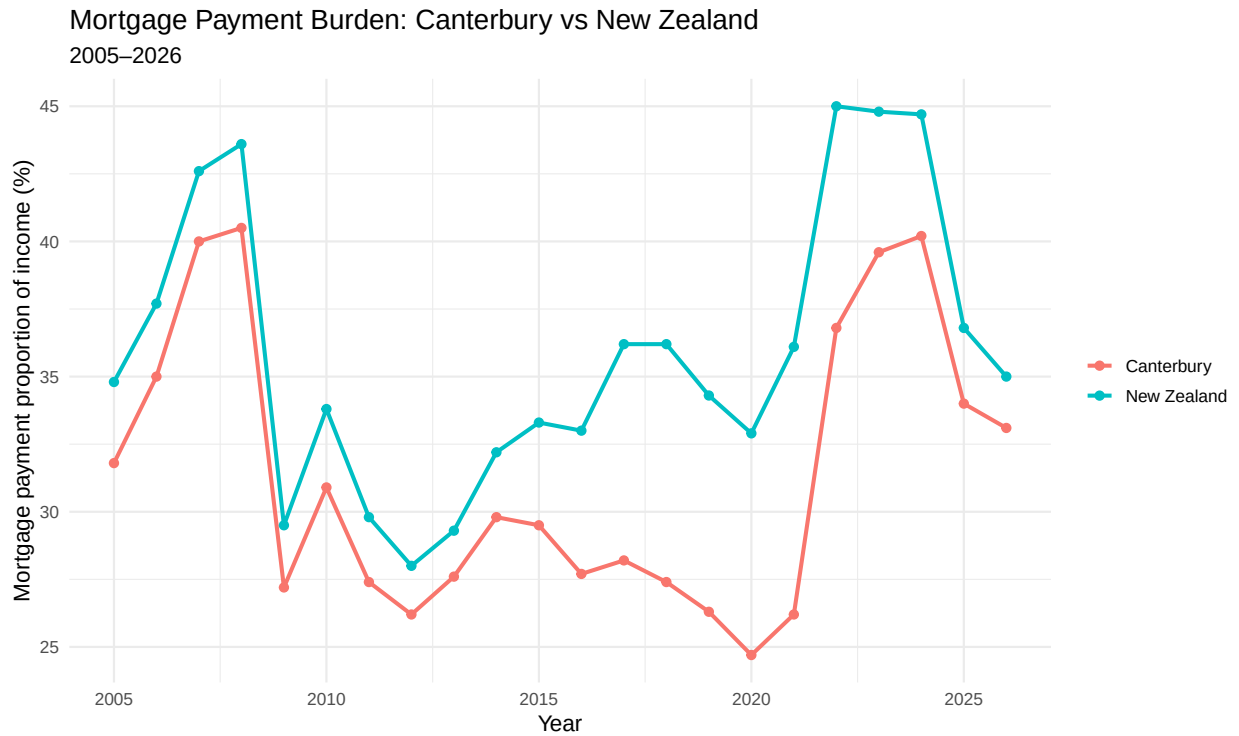
```



### 3.4 Mortgage-payment burden comparison

```
mortgage_long <- affordability_compare %>%
  select(
    Year,
    Canterbury =
      Canterbury_Mortgage,
    `New Zealand` =
      NZ_Mortgage
  ) %>%
  pivot_longer(
    -Year,
    names_to = "Region",
    values_to = "MortgageBurden"
  )

ggplot(
  mortgage_long,
  aes(
    x = Year,
    y = MortgageBurden,
    colour = Region
  )
) +
  geom_line(
    linewidth = 1.1
  ) +
  geom_point(
    size = 2
  ) +
  labs(
    title =
      "Mortgage Payment Burden: Canterbury vs New Zealand",
    subtitle =
      "2005-2026",
    x = "Year",
    y =
      "Mortgage payment proportion of income (%)",
    colour = NULL
  ) +
  theme_minimal(
    base_size = 13
  )
)
```



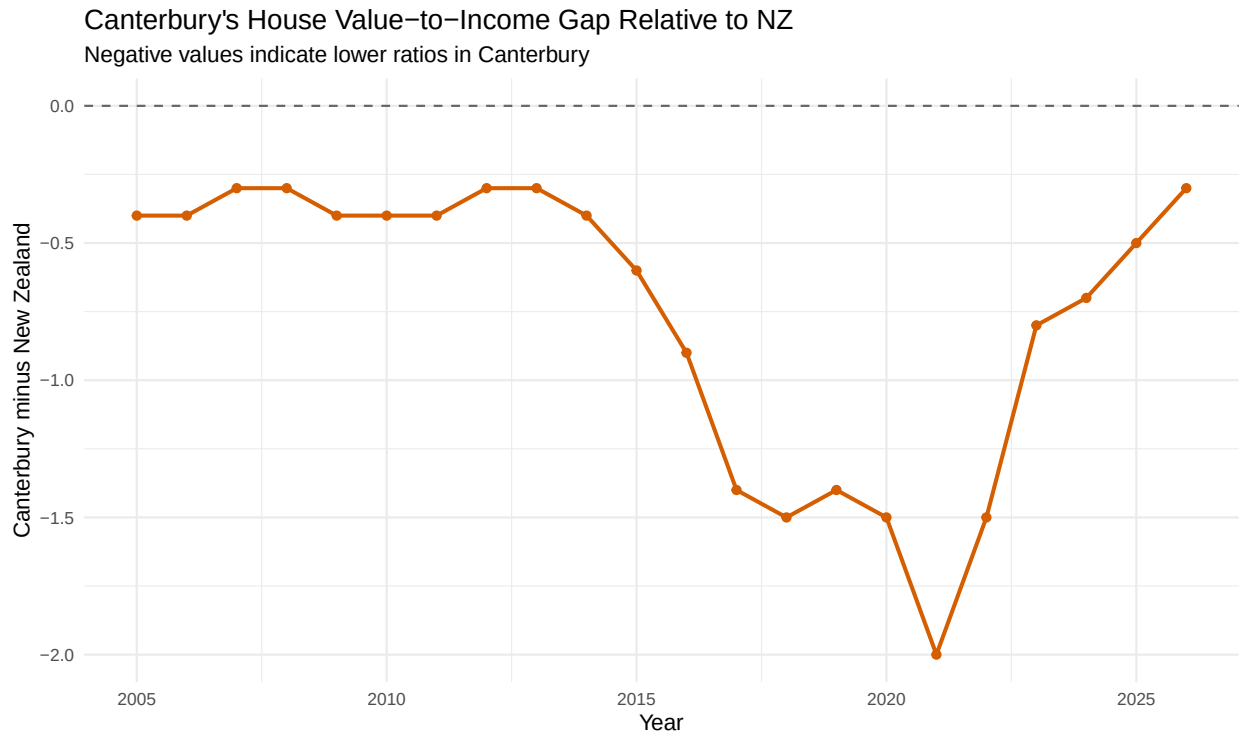
### 3.5 Affordability gap

```
ggplot(
  affordability_compare,
  aes(
    x = Year,
    y = ValueIncome_Gap
  )
) +
  geom_hline(
    yintercept = 0,
    linetype = "dashed",
    colour = "grey40"
  ) +
  geom_line(
    colour = "#D55E00",
    linewidth = 1.1
  ) +
  geom_point(
    colour = "#D55E00",
    size = 2
  ) +
  labs(
    title =
      "Canterbury's House Value-to-Income Gap Relative to NZ",
```

```

subtitle =
  "Negative values indicate lower ratios in Canterbury",
x = "Year",
y =
  "Canterbury minus New Zealand"
) +
theme_minimal(
  base_size = 13
)

```



### 3.6 Paired statistical tests

Because the Canterbury and New Zealand observations refer to the same years, paired tests are used.

```

value_test <- t.test(
  affordability_compare$Canterbury_ValueIncome,
  affordability_compare$NZ_ValueIncome,
  paired = TRUE
)

mortgage_test <- t.test(
  affordability_compare$Canterbury_Mortgage,
  affordability_compare$NZ_Mortgage,
  paired = TRUE
)

```

```
value_test
```

```
##
## Paired t-test
##
## data: affordability_compare$Canterbury_ValueIncome and affordability_compare$NZ_ValueIncome
## t = -6.6871, df = 21, p-value = 1.284e-06
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
## -0.9951585 -0.5230234
## sample estimates:
## mean difference
## -0.7590909
```

```
mortgage_test
```

```
##
## Paired t-test
##
## data: affordability_compare$Canterbury_Mortgage and affordability_compare$NZ_Mortgage
## t = -7.8541, df = 21, p-value = 1.105e-07
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
## -5.720265 -3.325190
## sample estimates:
## mean difference
## -4.522727
```

## Finding

The mean Canterbury house value-to-income multiple was approximately **0.76 lower than New Zealand**.

This difference was statistically significant:

**$p < 0.001$ .**

Canterbury's mortgage-payment burden was approximately **4.52 percentage points lower than New Zealand**, also with:

**$p < 0.001$ .**

Therefore, Canterbury has historically been significantly more affordable than New Zealand overall on both measures.

However, affordability still deteriorated considerably during the early 2020s.

## 4. Analysis 2 — What Is Associated with Canterbury House-Price Growth?

### 4.1 Prepare monthly housing-market data

```
analysis2 <- ts_raw %>%
  transmute(
    Date =
      as.Date(Date),

    Canterbury_Price =
      as.numeric(
        `Canterbury Median sale price`
      ),

    NZ_Price =
      as.numeric(
        `NZ Median sale price`
      ),

    Mortgage_1Y =
      as.numeric(
        `1 Year Mortgage Rate`
      ),

    Net_Migration =
      as.numeric(
        `Net migration`
      ),

    BrightLine =
      as.numeric(
        `Bright Line Test Period`
      ),

    Canterbury_Sales =
      as.numeric(
        `Volume Sold Canterbury`
      ),

    LVR_Total =
      as.numeric(
        `LVR Total New Commitments (NZDm)`
      ),

    Canterbury_Consents =
      as.numeric(
        `Canterbury Dwelling Unit Consents`
      ),

    Canterbury_Rent =
      as.numeric(
```

```

    `Canterbury Median Rent`
  ),

  NZ_Rent =
    as.numeric(
      `Median Rent`
    ),

  Canterbury_Income =
    as.numeric(
      `Mean Income Canterbury`
    ),

  NZ_Income =
    as.numeric(
      `Mean Income`
    )
) %>%
arrange(Date)

```

---

## 4.2 Create growth rates and lagged variables

Year-on-year changes are used to reduce the risk of misleading relationships caused purely by common upward trends.

```

analysis2 <- analysis2 %>%
  mutate(
    Year =
      as.integer(
        format(
          Date,
          "%Y"
        )
      ),

    Canterbury_Price_YoY =
      100 * (
        Canterbury_Price /
        lag(
          Canterbury_Price,
          12
        ) - 1
      ),

    NZ_Price_YoY =
      100 * (
        NZ_Price /
        lag(
          NZ_Price,
          12
        )
      )
  )

```

```

    ) - 1
  ),

  Canterbury_Sales_YoY =
    100 * (
      Canterbury_Sales /
      lag(
        Canterbury_Sales,
        12
      ) - 1
    ),

  LVR_Total_YoY =
    100 * (
      LVR_Total /
      lag(
        LVR_Total,
        12
      ) - 1
    ),

  Canterbury_Consents_YoY =
    100 * (
      Canterbury_Consents /
      lag(
        Canterbury_Consents,
        12
      ) - 1
    ),

  Migration_Change12 =
    Net_Migration -
    lag(
      Net_Migration,
      12
    ),

  Mortgage1Y_Lag6 =
    lag(
      Mortgage_1Y,
      6
    ),

  Migration_Lag6 =
    lag(
      Migration_Change12,
      6
    ),

  LVRGrowth_Lag6 =
    lag(
      LVR_Total_YoY,
      6

```

```

    ),

    Consents_Lag12 =
      lag(
        Canterbury_Consents_YoY,
        12
      ),

    SalesGrowth_Lag1 =
      lag(
        Canterbury_Sales_YoY,
        1
      )
  )
)

```

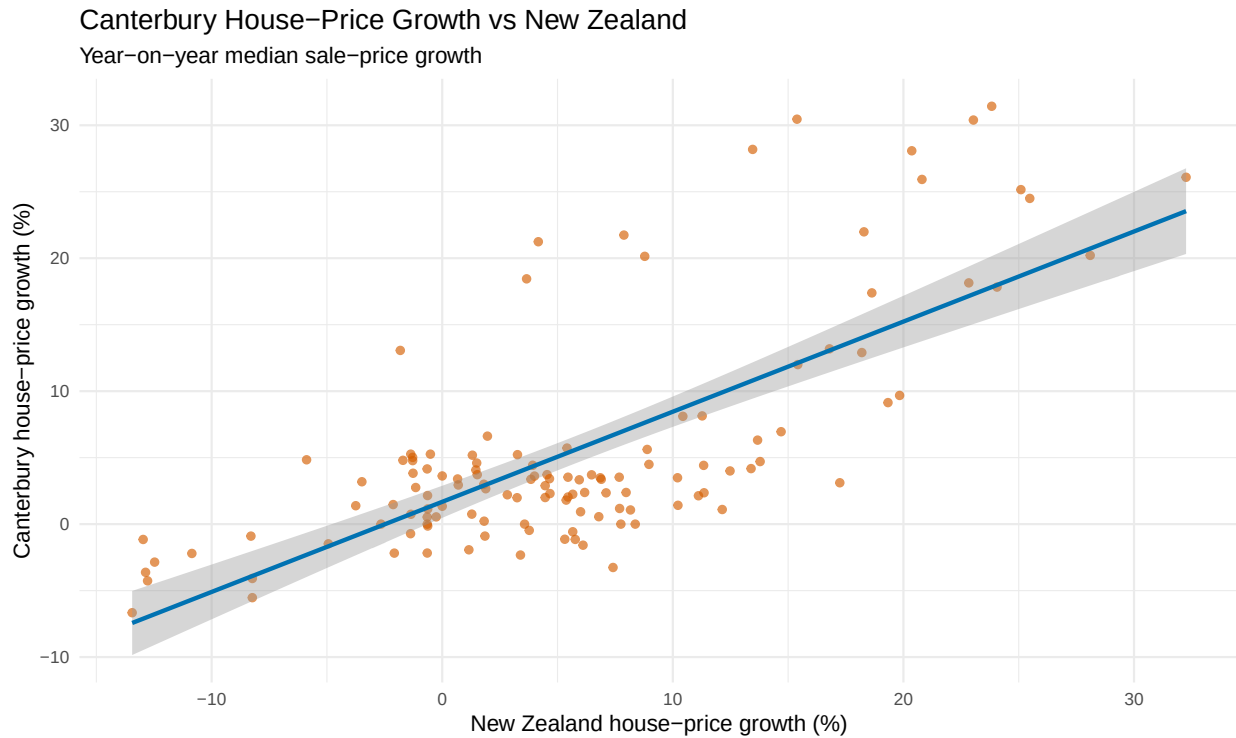
---

### 4.3 Canterbury vs national house-price growth

```

ggplot(
  analysis2,
  aes(
    x = NZ_Price_YoY,
    y = Canterbury_Price_YoY
  )
) +
  geom_point(
    alpha = 0.65,
    colour = "#D55E00"
  ) +
  geom_smooth(
    method = "lm",
    se = TRUE,
    colour = "#0072B2"
  ) +
  labs(
    title =
      "Canterbury House-Price Growth vs New Zealand",
    subtitle =
      "Year-on-year median sale-price growth",
    x =
      "New Zealand house-price growth (%)",
    y =
      "Canterbury house-price growth (%)"
  ) +
  theme_minimal(
    base_size = 13
  )

```



#### 4.4 Key regression models

Only the models that contribute directly to the final interpretation are retained.

```
model_nz <- lm(  
  Canterbury_Price_YoY ~  
    NZ_Price_YoY,  
  data = analysis2  
)
```

```
model_rate <- lm(  
  Canterbury_Price_YoY ~  
    Mortgage1Y_Lag6,  
  data = analysis2  
)
```

```
model_credit <- lm(  
  Canterbury_Price_YoY ~  
    LVRGrowth_Lag6,  
  data = analysis2  
)
```

```
main_model <- lm(  
  Canterbury_Price_YoY ~  
    NZ_Price_YoY +  
    Mortgage1Y_Lag6 +
```

```

    Migration_Lag6 +
    LVRGrowth_Lag6 +
    Consents_Lag12 +
    SalesGrowth_Lag1 +
    BrightLine,
    data = analysis2
  )

```

---

## 4.5 Model comparison

```

analysis2_models <- list(
  "NZ housing market" =
    model_nz,

  "Mortgage rate" =
    model_rate,

  "Credit growth" =
    model_credit,

  "Combined model" =
    main_model
)

model_comparison <- purrr::imap_dfr(
  analysis2_models,
  ~ broom::glance(.x) %>%
    mutate(
      Model = .y
    )
) %>%
  select(
    Model,
    r.squared,
    adj.r.squared,
    statistic,
    p.value,
    nobs,
    AIC,
    BIC
  )

kable(
  model_comparison,
  digits = 3,
  col.names = c(
    "Model",
    "R2",
    "Adjusted R2",

```

```

    "F-statistic",
    "p-value",
    "Observations",
    "AIC",
    "BIC"
  ),
  caption =
    "Canterbury House-Price Model Comparison"
)

```

Table 3: Canterbury House-Price Model Comparison

| Model             | R <sup>2</sup> | Adjusted R <sup>2</sup> | F-statistic | p-value | Observations | AIC     | BIC     |
|-------------------|----------------|-------------------------|-------------|---------|--------------|---------|---------|
| NZ housing market | 0.525          | 0.521                   | 136.945     | 0       | 126          | 804.649 | 813.157 |
| Mortgage rate     | 0.372          | 0.367                   | 73.607      | 0       | 126          | 839.680 | 848.189 |
| Credit growth     | 0.266          | 0.260                   | 42.782      | 0       | 120          | 824.017 | 832.380 |
| Combined model    | 0.823          | 0.811                   | 70.493      | 0       | 114          | 638.155 | 662.780 |

## 4.6 Newey-West robust inference

The housing time series contain serial correlation.

Therefore, Newey-West robust standard errors are used for the final statistical inference.

```

analysis2_robust <- bind_rows(

  nw_tidy(
    model_nz,
    "NZ housing market"
  ),

  nw_tidy(
    model_rate,
    "Mortgage rate"
  ),

  nw_tidy(
    model_credit,
    "Credit growth"
  ),

  nw_tidy(
    main_model,
    "Combined model"
  )
) %>%
mutate(
  term = dplyr::recode(

```

```

term,
"(Intercept)" =
  "Intercept",

"NZ_Price_YoY" =
  "NZ house-price growth",

"Mortgage1Y_Lag6" =
  "1-year mortgage rate (6m lag)",

"Migration_Lag6" =
  "Net migration change (6m lag)",

"LVRGrowth_Lag6" =
  "Credit growth (6m lag)",

"Consents_Lag12" =
  "Dwelling consents growth (12m lag)",

"SalesGrowth_Lag1" =
  "Sales growth (1m lag)",

"BrightLine" =
  "Bright-Line period"
)
)

kable(
  analysis2_robust,
  digits = 4,
  col.names = c(
    "Model",
    "Variable",
    "Estimate",
    "Robust SE",
    "t-statistic",
    "p-value"
  ),
  caption =
    "Newey-West Robust Results for Canterbury House-Price Growth"
)

```

Table 4: Newey-West Robust Results for Canterbury House-Price Growth

| Model             | Variable                      | Estimate | Robust SE | t-statistic | p-value |
|-------------------|-------------------------------|----------|-----------|-------------|---------|
| NZ housing market | Intercept                     | 1.6784   | 1.2137    | 1.3828      | 0.1692  |
| NZ housing market | NZ house-price growth         | 0.6778   | 0.1436    | 4.7184      | 0.0000  |
| Mortgage rate     | Intercept                     | 26.7376  | 10.2331   | 2.6129      | 0.0101  |
| Mortgage rate     | 1-year mortgage rate (6m lag) | -4.1000  | 1.8234    | -2.2485     | 0.0263  |

| Model          | Variable                           | Estimate | Robust SE | t-statistic | p-value |
|----------------|------------------------------------|----------|-----------|-------------|---------|
| Credit growth  | Intercept                          | 4.4988   | 1.7641    | 2.5502      | 0.0120  |
| Credit growth  | Credit growth (6m lag)             | 0.1392   | 0.0396    | 3.5192      | 0.0006  |
| Combined model | Intercept                          | -10.7137 | 5.4597    | -1.9623     | 0.0523  |
| Combined model | NZ house-price growth              | 0.7062   | 0.1113    | 6.3482      | 0.0000  |
| Combined model | 1-year mortgage rate (6m lag)      | 0.9566   | 0.8181    | 1.1693      | 0.2449  |
| Combined model | Net migration change (6m lag)      | -0.0001  | 0.0000    | -3.3335     | 0.0012  |
| Combined model | Credit growth (6m lag)             | 0.0301   | 0.0208    | 1.4421      | 0.1522  |
| Combined model | Dwelling consents growth (12m lag) | -0.0051  | 0.0134    | -0.3801     | 0.7046  |
| Combined model | Sales growth (1m lag)              | -0.0151  | 0.0102    | -1.4731     | 0.1437  |
| Combined model | Bright-Line period                 | 1.2875   | 0.2084    | 6.1770      | 0.0000  |

## Main results

The strongest and most consistent result was the relationship between Canterbury and the wider New Zealand housing market.

In the combined model, New Zealand house-price growth had an estimated coefficient of approximately:

**0.706**

with:

**p < 0.001.**

This means that, holding the other included variables constant, a one percentage-point increase in annual New Zealand house-price growth was associated with approximately **0.71 percentage points higher Canterbury house-price growth.**

The combined model produced approximately:

**R<sup>2</sup> = 0.823**

and:

**Adjusted R<sup>2</sup> = 0.811.**

Mortgage rates showed a significant negative relationship when analysed alone.

The standalone six-month-lagged mortgage-rate coefficient was approximately:

**-4.10**

with:

**p = 0.026.**

Credit growth also displayed a significant positive standalone association.

However, these relationships weakened after national housing conditions and the other variables were controlled for.

This suggests that Canterbury's housing market is strongly connected to the broader New Zealand housing cycle.

## 4.7 Final diagnostics

Only the diagnostics required to justify the final regression inference are retained.

```
bg_main <- bgtest(
  main_model,
  order = 12
)

vif_main <- car::vif(
  main_model
)

bp_main <- bptest(
  main_model
)

bg_main

##
## Breusch-Godfrey test for serial correlation of order up to 12
##
## data: main_model
## LM test = 42.09, df = 12, p-value = 3.216e-05

vif_main

##      NZ_Price_YoY Mortgage1Y_Lag6 Migration_Lag6 LVRGrowth_Lag6
##      4.583012      4.522626      2.325342      1.998617
## Consents_Lag12 SalesGrowth_Lag1      BrightLine
##      1.389129      1.403149      1.371546

bp_main

##
## studentized Breusch-Pagan test
##
## data: main_model
## BP = 9.4007, df = 7, p-value = 0.2252
```

### Diagnostic interpretation

The Breusch-Godfrey test identified significant residual serial correlation:

**$p < 0.001$ .**

This supports the use of Newey-West robust standard errors.

The largest VIF values were approximately **4.6**, indicating moderate but not severe multicollinearity.

The Breusch-Pagan test produced:

**$p = 0.225$ .**

Therefore, there was no significant evidence of heteroskedasticity in the combined model.

## 5. Analysis 3 — House Prices, Rents and Income

### 5.1 Purpose

This analysis examines whether household incomes and rents kept pace with Canterbury house prices. It connects housing-market movements more directly to household affordability.

---

### 5.2 Monthly data

```
analysis3_monthly <- analysis2 %>%
  select(
    Date,
    Year,
    Canterbury_Price,
    NZ_Price,
    Canterbury_Rent,
    NZ_Rent,
    Canterbury_Income,
    NZ_Income
  ) %>%
  mutate(
    Canterbury_Price_YoY =
      100 * (
        Canterbury_Price /
        lag(
          Canterbury_Price,
          12
        ) - 1
      ),

    Canterbury_Rent_YoY =
      100 * (
        Canterbury_Rent /
        lag(
          Canterbury_Rent,
          12
        ) - 1
      ),

    PriceGrowth_Lag12 =
      lag(
        Canterbury_Price_YoY,
        12
      ),

    Period =
      case_when(
        Year <= 2019 ~
          "2015-2019",
```

```

    Year <= 2022 ~
      "2020-2022",

    TRUE ~
      "2023 onwards"
  ),

  Period =
    factor(
      Period,
      levels = c(
        "2015-2019",
        "2020-2022",
        "2023 onwards"
      )
    )
  )
)

```

---

### 5.3 Annual prices, rents and income

Income is available annually rather than monthly.

Missing monthly values are not artificially interpolated.

```

analysis3_annual <- analysis3_monthly %>%
  group_by(
    Year
  ) %>%
  summarise(
    Canterbury_Price =
      median(
        Canterbury_Price,
        na.rm = TRUE
      ),

    NZ_Price =
      median(
        NZ_Price,
        na.rm = TRUE
      ),

    Canterbury_Rent =
      median(
        Canterbury_Rent,
        na.rm = TRUE
      ),

    NZ_Rent =
      median(
        NZ_Rent,

```

```

    na.rm = TRUE
  ),

  Canterbury_Income =
    first_non_missing(
      Canterbury_Income
    ),

  NZ_Income =
    first_non_missing(
      NZ_Income
    ),

  .groups = "drop"
)

```

---

## 5.4 Indexed house prices, rents and income

```

base_year <- min(
  analysis3_annual$Year,
  na.rm = TRUE
)

index_base <- analysis3_annual %>%
  filter(
    Year == base_year
  )

analysis3_index <- analysis3_annual %>%
  mutate(
    House_Prices =
      100 *
      Canterbury_Price /
      index_base$Canterbury_Price,

    Rents =
      100 *
      Canterbury_Rent /
      index_base$Canterbury_Rent,

    Income =
      100 *
      Canterbury_Income /
      index_base$Canterbury_Income
  ) %>%
  select(
    Year,
    House_Prices,
    Rents,

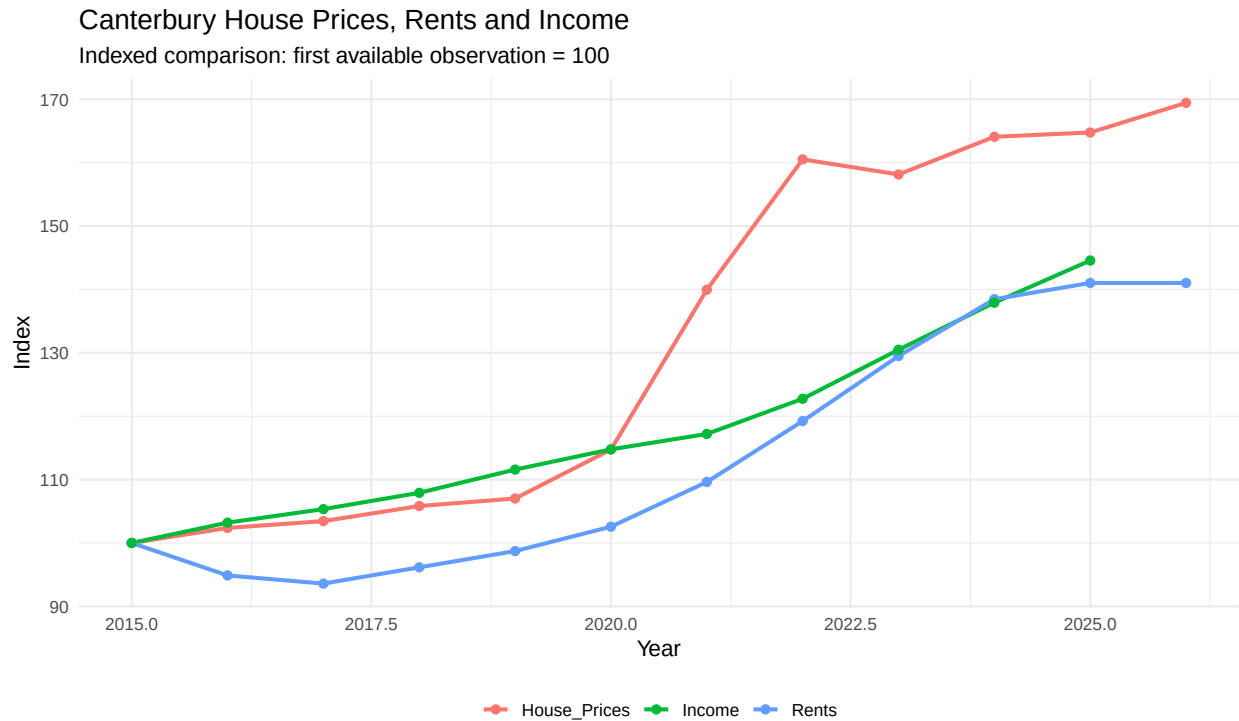
```

```

    Income
  ) %>%
  pivot_longer(
    -Year,
    names_to = "Measure",
    values_to = "Index"
  )

ggplot(
  analysis3_index,
  aes(
    x = Year,
    y = Index,
    colour = Measure
  )
) +
  geom_line(
    linewidth = 1.1,
    na.rm = TRUE
  ) +
  geom_point(
    size = 2,
    na.rm = TRUE
  ) +
  labs(
    title =
      "Canterbury House Prices, Rents and Income",
    subtitle =
      "Indexed comparison: first available observation = 100",
    x = "Year",
    y = "Index",
    colour = NULL
  ) +
  theme_minimal(
    base_size = 13
  ) +
  theme(
    legend.position = "bottom"
  )

```



## 5.5 House-price-to-income proxy

This is a separate measure from the official house value-to-income multiple used in Analysis 1.

```
price_income_table <- analysis3_annual %>%
  filter(
    !is.na(
      Canterbury_Income
    ),
    !is.na(
      NZ_Income
    )
  ) %>%
  mutate(
    Canterbury_PriceIncome =
      Canterbury_Price /
      Canterbury_Income,

    NZ_PriceIncome =
      NZ_Price /
      NZ_Income
  )

kable(
  price_income_table %>%
  select(
```

```

    Year,
    Canterbury_Price,
    Canterbury_Income,
    Canterbury_PriceIncome,
    NZ_Price,
    NZ_Income,
    NZ_PriceIncome
  ),
  digits = 2,
  col.names = c(
    "Year",
    "Canterbury House Price ($)",
    "Canterbury Income ($)",
    "Canterbury Price-to-Income",
    "NZ House Price ($)",
    "NZ Income ($)",
    "NZ Price-to-Income"
  ),
  caption =
    "House-Price-to-Income Proxy"
)

```

Table 5: House-Price-to-Income Proxy

| Year | Canterbury House<br>Price<br>( $ CanterburyIncome()$ ) | Canterbury<br>Price-to-<br>Income | NZ House Price<br>( $ NZIncome()$ ) | NZ Price-to-<br>Income |       |       |
|------|--------------------------------------------------------|-----------------------------------|-------------------------------------|------------------------|-------|-------|
| 2015 | 420500.0                                               | 53228                             | 7.90                                | 459000                 | 55450 | 8.28  |
| 2016 | 430463.5                                               | 54932                             | 7.84                                | 500500                 | 56995 | 8.78  |
| 2017 | 435000.0                                               | 56068                             | 7.76                                | 530000                 | 58418 | 9.07  |
| 2018 | 445000.0                                               | 57440                             | 7.75                                | 559500                 | 59830 | 9.35  |
| 2019 | 450000.0                                               | 59390                             | 7.58                                | 582500                 | 62000 | 9.40  |
| 2020 | 482500.0                                               | 61088                             | 7.90                                | 671200                 | 64170 | 10.46 |
| 2021 | 588500.0                                               | 62382                             | 9.43                                | 822500                 | 65910 | 12.48 |
| 2022 | 675000.0                                               | 65338                             | 10.33                               | 834000                 | 69620 | 11.98 |
| 2023 | 665000.0                                               | 69455                             | 9.57                                | 780000                 | 74395 | 10.48 |
| 2024 | 690000.0                                               | 73410                             | 9.40                                | 779000                 | 78648 | 9.90  |
| 2025 | 692826.0                                               | 76950                             | 9.00                                | 771500                 | 81958 | 9.41  |

## 5.6 Canterbury price-to-income trend

```

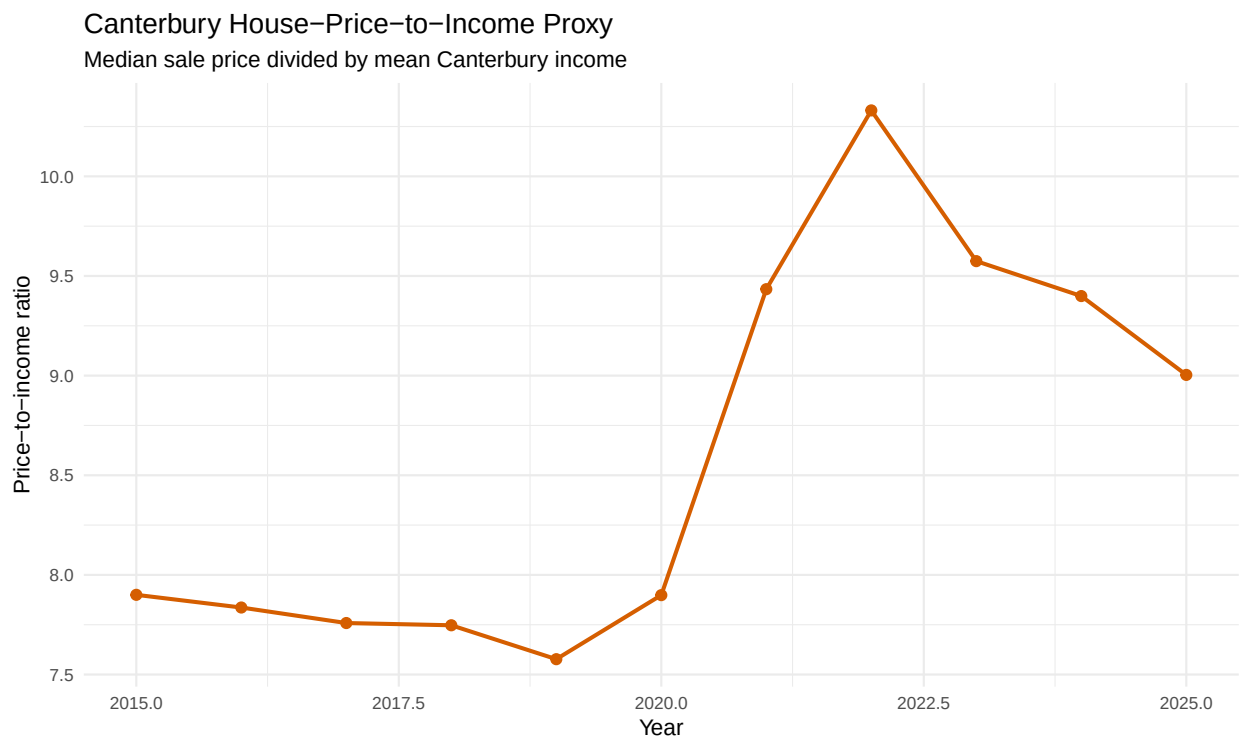
ggplot(
  price_income_table,
  aes(
    x = Year,
    y = Canterbury_PriceIncome
  )
) +

```

```

geom_line(
  colour = "#D55E00",
  linewidth = 1.1
) +
geom_point(
  colour = "#D55E00",
  size = 2.5
) +
labs(
  title =
    "Canterbury House-Price-to-Income Proxy",
  subtitle =
    "Median sale price divided by mean Canterbury income",
  x = "Year",
  y =
    "Price-to-income ratio"
) +
theme_minimal(
  base_size = 13
)

```



### Interpretation

The Canterbury price-to-income proxy increased from approximately **7.90 in 2020** to approximately **10.33 in 2022**.

It subsequently declined to approximately **9.00 by 2025**.

This indicates that purchase affordability deteriorated sharply during the 2020–2022 housing boom before partially improving.

---

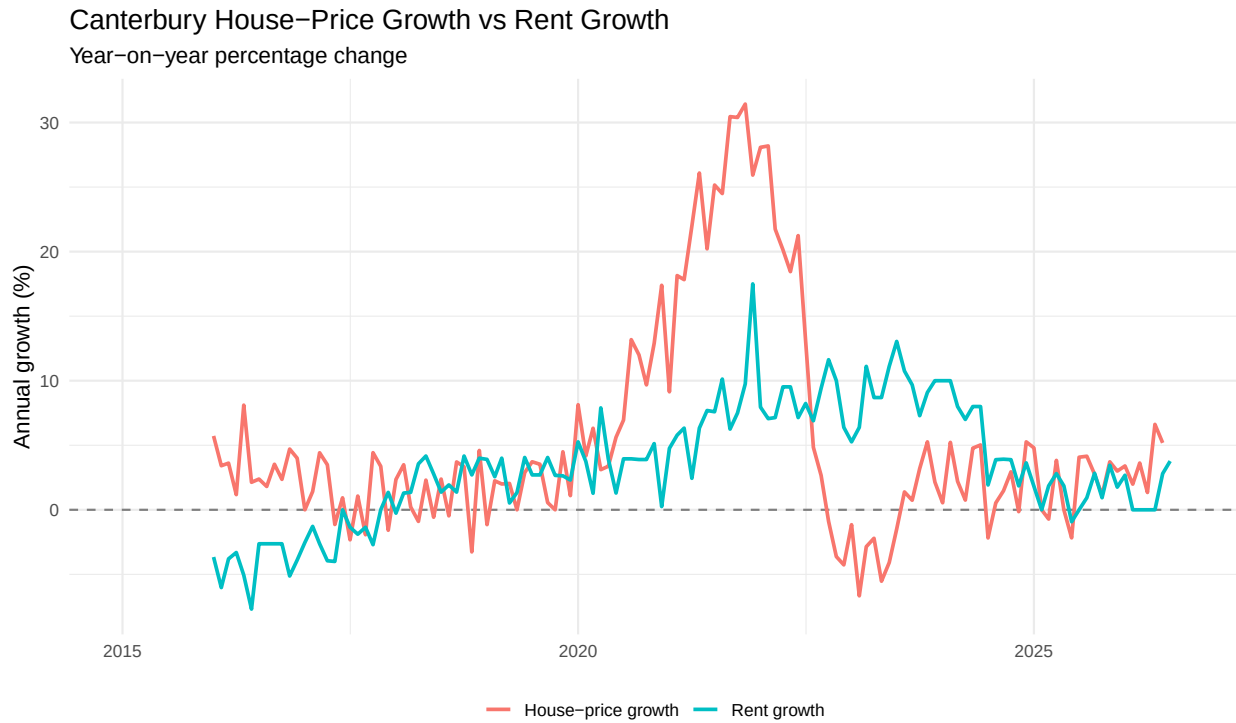
## 5.7 House-price growth vs rent growth

```
price_rent_growth <- analysis3_monthly %>%
  select(
    Date,

    `House-price growth` =
      Canterbury_Price_YoY,

    `Rent growth` =
      Canterbury_Rent_YoY
  ) %>%
  pivot_longer(
    -Date,
    names_to = "Measure",
    values_to = "Growth"
  )

ggplot(
  price_rent_growth,
  aes(
    x = Date,
    y = Growth,
    colour = Measure
  )
) +
  geom_hline(
    yintercept = 0,
    linetype = "dashed",
    colour = "grey50"
  ) +
  geom_line(
    linewidth = 1
  ) +
  labs(
    title =
      "Canterbury House-Price Growth vs Rent Growth",
    subtitle =
      "Year-on-year percentage change",
    x = NULL,
    y = "Annual growth (%)",
    colour = NULL
  ) +
  theme_minimal(
    base_size = 13
  ) +
  theme(
    legend.position = "bottom"
  )
```



## 5.8 Housing-market periods

```
period_summary <- analysis3_monthly %>%
  group_by(
    Period
  ) %>%
  summarise(
    Mean_Price_Growth =
      mean(
        Canterbury_Price_YoY,
        na.rm = TRUE
      ),

    Mean_Rent_Growth =
      mean(
        Canterbury_Rent_YoY,
        na.rm = TRUE
      ),

    .groups = "drop"
  )

kable(
  period_summary,
  digits = 2,
```

```
col.names = c(
  "Period",
  "House-Price Growth (%)",
  "Rent Growth (%)")
),
caption =
  "Canterbury Housing Growth by Period"
)
```

Table 6: Canterbury Housing Growth by Period

| Period       | House-Price Growth (%) | Rent Growth (%) |
|--------------|------------------------|-----------------|
| 2015-2019    | 1.95                   | -0.16           |
| 2020-2022    | 14.82                  | 6.60            |
| 2023 onwards | 1.47                   | 4.83            |

## Interpretation

Average Canterbury house-price growth was approximately:

- **1.95% during 2015–2019**
- **14.82% during 2020–2022**
- **1.47% from 2023 onward**

Average rent growth was approximately:

- **-0.16% during 2015–2019**
- **6.60% during 2020–2022**
- **4.83% from 2023 onward**

The nature of affordability pressure therefore changed over time.

During 2020–2022, rapidly rising purchase prices were the dominant problem.

After the housing boom, house-price growth slowed substantially while rental growth remained elevated.

## 5.9 Rent-growth models

To keep the final analysis concise, only the immediate relationship and the strongest individual lag relationship are retained.

```
rent_current_model <- lm(
  Canterbury_Rent_YoY ~
    Canterbury_Price_YoY,
  data = analysis3_monthly
)

rent_lag12_model <- lm(
```

```

Canterbury_Rent_YoY ~
  PriceGrowth_Lag12,
data = analysis3_monthly
)

```

---

## 5.10 Newey-West rent results

```

analysis3_robust <- bind_rows(

  nw_tidy(
    rent_current_model,
    "Current relationship"
  ),

  nw_tidy(
    rent_lag12_model,
    "12-month lag"
  )
) %>%
mutate(
  term = dplyr::recode(
    term,
    "(Intercept)" =
      "Intercept",

    "Canterbury_Price_YoY" =
      "Current house-price growth",

    "PriceGrowth_Lag12" =
      "House-price growth (12m lag)"
  )
)

kable(
  analysis3_robust,
  digits = 4,
  col.names = c(
    "Model",
    "Variable",
    "Estimate",
    "Robust SE",
    "t-statistic",
    "p-value"
  ),
  caption =
    "Newey-West Results for Canterbury Rent Growth"
)

```

Table 7: Newey-West Results for Canterbury Rent Growth

| Model                | Variable                     | Estimate | Robust<br>SE | t-statistic | p-value |
|----------------------|------------------------------|----------|--------------|-------------|---------|
| Current relationship | Intercept                    | 2.5306   | 1.5446       | 1.6383      | 0.1039  |
| Current relationship | Current house-price growth   | 0.1670   | 0.0873       | 1.9126      | 0.0581  |
| 12-month lag         | Intercept                    | 3.0019   | 1.1035       | 2.7204      | 0.0076  |
| 12-month lag         | House-price growth (12m lag) | 0.2170   | 0.0604       | 3.5930      | 0.0005  |

### Interpretation

The same-period relationship between house-price growth and rent growth was positive but only weakly significant after correcting for serial correlation.

The 12-month lag produced a stronger result.

The estimated 12-month coefficient was approximately:

**0.217**

with:

**p = 0.0005.**

This suggests that a **10 percentage-point increase in annual house-price growth was associated with approximately 2.17 percentage points higher annual rent growth one year later.**

This relationship should not be interpreted as proof that rising house prices directly cause higher rents.

---

## 6. Overall Findings

The analysis produces three main findings.

First, Canterbury has historically been more affordable than New Zealand overall.

Both Canterbury's house value-to-income multiple and mortgage-payment burden were significantly lower than the corresponding New Zealand measures.

Second, Canterbury's housing market remains strongly connected to the national housing cycle.

National house-price growth was the strongest and most consistent predictor in the final Canterbury house-price model.

Mortgage rates and credit conditions displayed meaningful standalone relationships with Canterbury price growth, but these relationships weakened once broader national housing conditions were controlled for.

Third, affordability deteriorated sharply during the 2020–2022 housing boom.

House prices increased substantially faster than household income.

Although purchase-price growth later slowed, rental growth remained relatively elevated.

The regression evidence also suggests that rent growth may respond to earlier housing-market conditions with a delay.

---

## 7. Limitations

The analysis has several limitations.

The data are observational, so the statistical relationships cannot be interpreted as causal effects.

Some explanatory variables, including migration and monetary-policy variables, are measured nationally rather than specifically for Canterbury.

Time-series models displayed serial correlation.

Newey-West robust standard errors were therefore used for final statistical inference.

Moderate multicollinearity was present between some explanatory variables, although VIF values remained below approximately five.

The Bright-Line variable represents different policy regimes over time and should not be interpreted as an isolated causal estimate of the policy's effect.

Dwelling consents represent approvals rather than completed housing supply.

Finally, Canterbury income is available at a lower frequency than the monthly housing-market measures.

Income was therefore analysed annually rather than artificially filling missing monthly observations.

---

## 8. Conclusion

Canterbury has historically maintained a housing-affordability advantage relative to New Zealand overall.

However, Canterbury still experienced substantial affordability deterioration during the early-2020s housing boom.

House prices increased much faster than household income during 2020–2022.

Although house-price growth subsequently slowed, rental costs continued to rise relatively strongly.

Canterbury house-price movements are closely connected to the wider New Zealand housing market, while borrowing conditions, credit availability, migration, housing supply and market activity provide additional context.

Overall, Canterbury housing affordability appears to reflect a combination of national housing-market conditions, household income, financing costs, housing supply and rental-market pressures rather than a single explanatory factor. ““